

ECON 320 — LECTURE NOTES

Exponential and Logarithmic Functions

Chiang & Wainwright (2005) Ch. 10 | Hoy et al. (2011) | Simon & Blume (1994)

Course ECON 320: Mathematical Economics III

Primary Text A. C. Chiang & K. Wainwright, *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw-Hill, 2005. Ch. 10.

Supplementary Hoy, Livernois, McKenna, Rees & Stengos, *Mathematics for Economics*, 3rd ed., MIT Press, 2011. Ch. 6.

Simon & Blume, *Mathematics for Economists*, Norton, 1994. Ch. 25.

Topics Exponential functions, the number e , logarithmic functions, laws of exponents and logarithms, natural logarithm, differentiation and integration, economic applications: compound interest, continuous growth, elasticity, log-linearisation.

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1 Introduction: Why Exponential and Log Functions in Economics?

Exponential and logarithmic functions are indispensable in economic analysis. Their ubiquity stems from a single, fundamental observation: *many economic quantities grow (or decay) at a rate proportional to their current level*. When a quantity's rate of change is proportional to the quantity itself, the law of motion is an exponential function. Examples include:

- **Capital accumulation:** $\dot{K} = sY$ leads to exponential growth paths in neoclassical growth models.
- **Compound interest:** A principal P at rate r grows as $P(1+r)^t$, approaching Pe^{rt} under continuous compounding.
- **Price dynamics:** Inflation at a constant rate π implies price level $P_t = P_0e^{\pi t}$.
- **Discounting:** The present value of a future payment uses e^{-rt} under continuous discounting.
- **Econometrics:** Log-linear specifications yield constant elasticities, simplify multiplicative models, and appear in almost every estimated demand and production equation.

Moreover, the natural logarithm transforms multiplicative relationships into additive ones, converts power functions into linear ones, and provides the cleanest measure of proportional change. For an economist, $\ln$ is as essential a tool as the derivative itself.

2 Exponential Functions

2.1 Definition and Basic Properties

Definition Exponential Function

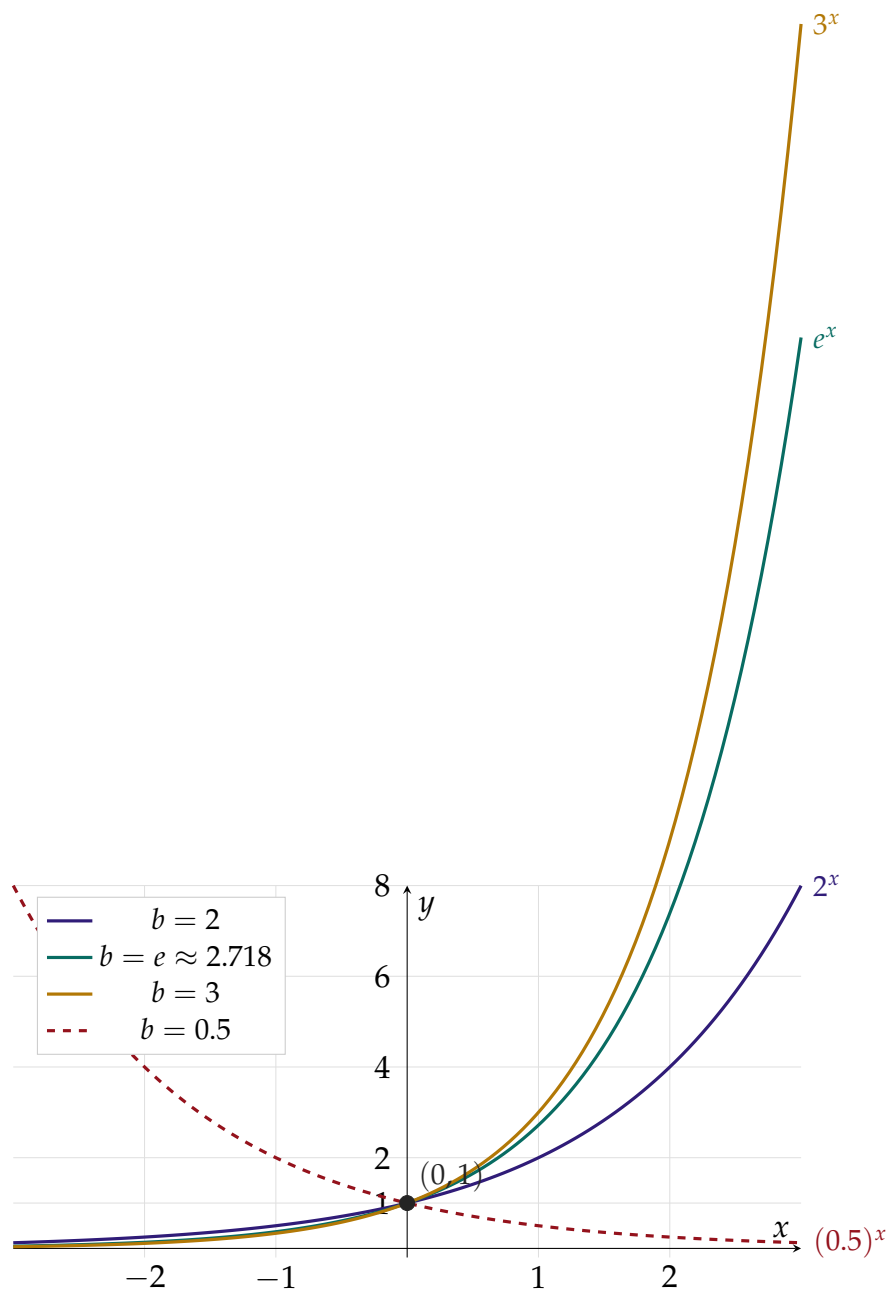
An **exponential function** has the form:

$$f(x) = b^x, \quad b > 0, b \neq 1, x \in \mathbb{R}$$

where b is the **base** and x is the **exponent**. The domain is all of $\mathbb{R}$ and the range is $(0, \infty)$.

The behaviour depends critically on the base:

- $b > 1$: $f(x) = b^x$ is **strictly increasing** (exponential growth). As $x \rightarrow \infty$, $f(x) \rightarrow \infty$; as $x \rightarrow -\infty$, $f(x) \rightarrow 0$.
- $0 < b < 1$: $f(x) = b^x$ is **strictly decreasing** (exponential decay). As $x \rightarrow \infty$, $f(x) \rightarrow 0$; as $x \rightarrow -\infty$, $f(x) \rightarrow \infty$.
- In all cases: $f(0) = b^0 = 1$, so every exponential function passes through $(0, 1)$.



All exponential functions pass through $(0, 1)$. Larger bases grow faster; bases less than 1 decay. The natural base e occupies a special position.

2.2 Laws of Exponents

Theorem Laws of Exponents

For $b > 0$, $c > 0$, and all $x, y \in \mathbb{R}$:

Law	Statement	Example
Product rule	$b^x \cdot b^y = b^{x+y}$	$2^3 \cdot 2^4 = 2^7 = 128$
Quotient rule	$b^x / b^y = b^{x-y}$	$5^6 / 5^2 = 5^4 = 625$
Power rule	$(b^x)^y = b^{xy}$	$(3^2)^4 = 3^8 = 6561$
Base product	$(bc)^x = b^x c^x$	$(2 \cdot 3)^4 = 2^4 \cdot 3^4 = 1296$
Base quotient	$(b/c)^x = b^x / c^x$	$(4/2)^3 = 4^3 / 2^3 = 8$
Zero exponent	$b^0 = 1$	$7^0 = 1$
Negative exponent	$b^{-x} = 1/b^x$	$2^{-3} = 1/8$
Fractional exponent	$b^{p/q} = (b^{1/q})^p = \sqrt[q]{b^p}$	$8^{2/3} = (\sqrt[3]{8})^2 = 4$

(Chiang & Wainwright, p. 268; Simon & Blume, Ch. 25)

Worked Example Simplifying Exponential Expressions

1. Simplify $\frac{(2x^3)^4 \cdot x^{-2}}{8x^5}$:

$$\frac{16x^{12} \cdot x^{-2}}{8x^5} = \frac{16x^{10}}{8x^5} = 2x^5$$

2. Solve for x : $4^x = 8$.

$$(2^2)^x = 2^3 \implies 2^{2x} = 2^3 \implies 2x = 3 \implies x = \frac{3}{2}$$

3. In economics, the Cobb-Douglas production function $Y = AK^\alpha L^\beta$ uses fractional exponents. Here $K^{0.3}$ means $K^{3/10} = \sqrt[10]{K^3}$, and the law $(tK)^\alpha (tL)^\beta = t^{\alpha+\beta} K^\alpha L^\beta$ follows directly from the base-product rule.

2.3 The Number e and the Natural Exponential

The most important base in mathematics and economics is the irrational number $e \approx 2.71828 \dots$, known as **Euler's number**.

Definition The Number e

The number e is defined as:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{h \rightarrow 0} (1 + h)^{1/h} \approx 2.71828182845 \dots$$

It also equals the sum of the series:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \dots$$

The function $f(x) = e^x$ is called the **natural exponential function**.

Why is e Special?

The number e is the unique base for which:

1. $\frac{d}{dx}[e^x] = e^x$ — the natural exponential is its own derivative.
2. $\int e^x dx = e^x + C$.
3. The tangent line to $y = e^x$ at $(0, 1)$ has slope exactly 1.
4. The solution to $\dot{y} = ky$ (proportional growth) is $y(t) = y_0 e^{kt}$.

No other base has property (1), which makes e the “natural” choice for calculus and differential equations. (*Chiang & Wainwright, p. 271*)

Worked Example Deriving e from Compound Interest

A bank offers annual interest rate r . If interest is compounded n times per year, \$1 invested grows to $\left(1 + \frac{r}{n}\right)^n$ after one year.

n (compoundings/yr)	Formula	Value at $r = 1$ (100%)
1 (annual)	$(1 + 1)^1$	2.000
2 (semi-annual)	$(1 + 0.5)^2$	2.250
4 (quarterly)	$(1 + 0.25)^4$	2.441
12 (monthly)	$(1 + 1/12)^{12}$	2.613
365 (daily)	$(1 + 1/365)^{365}$	2.715
∞ (continuous)	$\lim_{n \rightarrow \infty} (1 + 1/n)^n$	$e = 2.718 \dots$

As $n \rightarrow \infty$, the limit is e^r . **Under continuous compounding at rate r** , \$1 grows to e^r in one year and to e^{rt} in t years.

This derivation shows that e emerges naturally from the mathematics of interest, making it the canonical base for financial and economic modelling.

3 Logarithmic Functions

3.1 Definition as the Inverse of the Exponential

Definition Logarithm

The **logarithm base** b of a positive number y is the exponent x such that $b^x = y$:

$$x = \log_b y \iff b^x = y, \quad b > 0, b \neq 1, y > 0$$

The logarithm is the **inverse function** of the exponential: $\log_b(b^x) = x$ for all $x \in \mathbb{R}$, and $b^{\log_b y} = y$ for all $y > 0$.

The domain of $\log_b$ is $(0, \infty)$ and the range is $\mathbb{R}$.

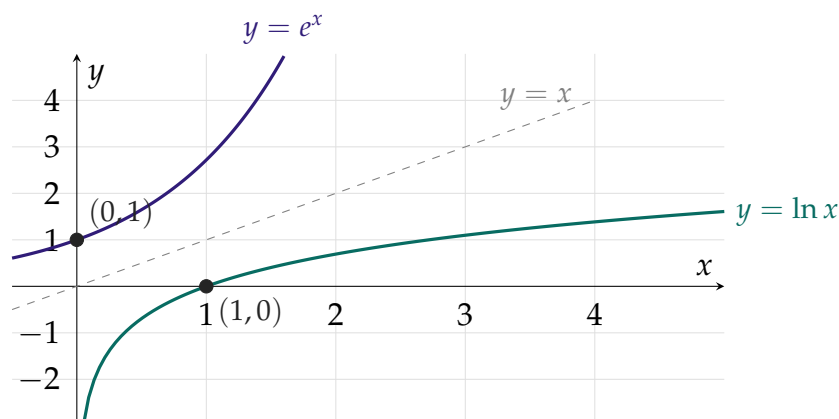
Definition Natural Logarithm

The **natural logarithm** is the logarithm base e :

$$\ln y = \log_e y \iff e^x = y$$

It is the inverse of the natural exponential: $\ln(e^x) = x$ and $e^{\ln y} = y$.

The natural log is the standard logarithm in economics and calculus because of its clean derivative and integral properties. (Chiang & Wainwright, p. 275)



$y = e^x$ and $y = \ln x$ are reflections of each other across the line $y = x$, confirming they are inverse functions. Note $e^0 = 1 \iff \ln 1 = 0$; $e^1 = e \iff \ln e = 1$.

3.2 Laws of Logarithms**Theorem Laws of Logarithms**

For $b > 0$, $b \neq 1$, and positive numbers M , N :

$$\log_b(MN) = \log_b M + \log_b N \quad (\text{Product})$$

$$\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N \quad (\text{Quotient})$$

$$\log_b(M^p) = p \log_b M, \quad p \in \mathbb{R} \quad (\text{Power})$$

$$\log_b b = 1 \quad (\text{Base})$$

$$\log_b 1 = 0 \quad (\text{Unity})$$

$$\log_b M = \frac{\ln M}{\ln b} = \frac{\log_a M}{\log_a b} \quad (\text{Change of Base})$$

In particular, writing these for the natural log ($b = e$):

$$\ln(MN) = \ln M + \ln N, \quad \ln\left(\frac{M}{N}\right) = \ln M - \ln N, \quad \ln(M^p) = p \ln M$$

Proof of the Product Law. Let $x = \log_b M$ and $y = \log_b N$, so $M = b^x$ and $N = b^y$. Then $MN = b^x \cdot b^y = b^{x+y}$ by the product rule for exponents. Therefore $\log_b(MN) =$

$$x + y = \log_b M + \log_b N.$$

□

□

Worked Example Applying the Laws of Logarithms

1. Expand $\ln\left(\frac{x^2\sqrt{y}}{z^3}\right)$:

$$\ln\left(\frac{x^2 y^{1/2}}{z^3}\right) = \ln(x^2) + \ln(y^{1/2}) - \ln(z^3) = 2\ln x + \frac{1}{2}\ln y - 3\ln z$$

2. Solve $2\ln x + \ln(x+1) = \ln 12$:

$$\ln(x^2) + \ln(x+1) = \ln 12 \implies \ln[x^2(x+1)] = \ln 12 \implies x^2(x+1) = 12 \implies x^3 + x^2 - 12 = 0$$

Factor: $(x-2)(x^2+3x+6) = 0$. Check: $x^3 + x^2 - 12$ at $x = 2$: $8 + 4 - 12 = 0$.
✓

So $x = 2$ (the quadratic $x^2 + 3x + 6$ has no real roots since discriminant $= 9 - 24 < 0$).

3. **Log-linearising Cobb-Douglas:** $Y = AK^\alpha L^\beta$. Taking logs:

$$\ln Y = \ln A + \alpha \ln K + \beta \ln L$$

This converts the multiplicative model into a linear one, enabling OLS regression. The coefficient α is directly the elasticity of output with respect to capital, and β is the labour elasticity.

4. Change of base: $\log_8 32$.

$$\log_8 32 = \frac{\ln 32}{\ln 8} = \frac{\ln 2^5}{\ln 2^3} = \frac{5 \ln 2}{3 \ln 2} = \frac{5}{3}$$

3.3 Relationship Between e and $\ln$

Key Identities: e and $\ln$

$$\ln e = 1$$

$$e^0 = 1 \implies \ln 1 = 0$$

$$\ln(e^x) = x \quad \forall x \in \mathbb{R}$$

$$e^{\ln y} = y \quad \forall y > 0$$

$$\ln e^r = r$$

$$e^{\ln a + \ln b} = ab$$

$$\ln(1+r) \approx r \text{ for small } r$$

$$e^r \approx 1+r \text{ for small } r$$

The approximations $\ln(1+r) \approx r$ and $e^r \approx 1+r$ are widely used for small growth rates r (e.g. quarterly inflation or interest rates).

4 Differentiation of Exponential and Logarithmic Functions

4.1 Derivatives of Exponential Functions

Theorem Derivatives of Exponential Functions

$$\frac{d}{dx}[e^x] = e^x \quad (1)$$

$$\frac{d}{dx}[e^{f(x)}] = e^{f(x)} \cdot f'(x) \quad (\text{Chain Rule}) \quad (2)$$

$$\frac{d}{dx}[a^x] = a^x \ln a \quad (a > 0, a \neq 1) \quad (3)$$

$$\frac{d}{dx}[a^{f(x)}] = a^{f(x)} \cdot f'(x) \cdot \ln a \quad (4)$$

(Chiang & Wainwright, p. 277; Hoy et al., p. 175)

Derivation of $d(e^x)/dx = e^x$. From first principles:

$$\frac{d}{dx}[e^x] = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = \lim_{h \rightarrow 0} e^x \cdot \frac{e^h - 1}{h} = e^x \cdot \underbrace{\lim_{h \rightarrow 0} \frac{e^h - 1}{h}}_{=1} = e^x$$

The key step uses the standard limit $\lim_{h \rightarrow 0} (e^h - 1)/h = 1$, which follows from the definition of e .

For $a^x = e^{x \ln a}$ (writing $a = e^{\ln a}$), apply the chain rule: $\frac{d}{dx}[e^{x \ln a}] = e^{x \ln a} \cdot \ln a = a^x \ln a$. □

Worked Example Differentiating Exponential Functions

1. $f(x) = e^{3x^2-5}$:

$$f'(x) = e^{3x^2-5} \cdot 6x = 6xe^{3x^2-5}$$

2. $g(t) = 5e^{-0.08t}$ (present-value factor with continuous discounting at 8%):

$$g'(t) = 5e^{-0.08t} \cdot (-0.08) = -0.4e^{-0.08t}$$

Interpretation: the discount factor is decaying at 8% per period.

3. $h(x) = x^2 e^{-x}$. Product rule with $u = x^2$, $v = e^{-x}$:

$$h'(x) = 2xe^{-x} + x^2(-e^{-x}) = e^{-x}(2x - x^2) = xe^{-x}(2 - x)$$

Critical points: $x = 0$ (min) and $x = 2$ (max, since $h''(2) < 0$).

4. $k(x) = 2^{3x}$:

$$k'(x) = 2^{3x} \cdot \ln 2 \cdot 3 = 3 \ln(2) \cdot 2^{3x}$$

4.2 Derivatives of Logarithmic Functions

Theorem Derivatives of Logarithmic Functions

$$\frac{d}{dx}[\ln x] = \frac{1}{x}, \quad x > 0 \quad (5)$$

$$\frac{d}{dx}[\ln |x|] = \frac{1}{x}, \quad x \neq 0 \quad (6)$$

$$\frac{d}{dx}[\ln f(x)] = \frac{f'(x)}{f(x)}, \quad f(x) > 0 \quad (7)$$

$$\frac{d}{dx}[\log_a x] = \frac{1}{x \ln a}, \quad x > 0, a > 0, a \neq 1 \quad (8)$$

(Chiang & Wainwright, p. 278; Simon & Blume, p. 540)

Derivation of $d(\ln x)/dx = 1/x$. Let $y = \ln x$, so $e^y = x$. Differentiating implicitly with respect to x :

$$e^y \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}. \quad \square$$

Alternatively, by the Inverse Function Rule: since $\frac{d}{dy}[e^y] = e^y$, $\frac{dx}{dy} = e^y$, so $\frac{dy}{dx} = 1/e^y = 1/x$. $\square$

Worked Example Differentiating Logarithmic Functions

1. $f(x) = \ln(3x^2 + 1)$:

$$f'(x) = \frac{6x}{3x^2 + 1}$$

2. $g(x) = \ln\left(\frac{x^2 + 1}{x - 1}\right) = \ln(x^2 + 1) - \ln(x - 1)$, $x > 1$:

$$g'(x) = \frac{2x}{x^2 + 1} - \frac{1}{x - 1} = \frac{2x(x - 1) - (x^2 + 1)}{(x^2 + 1)(x - 1)} = \frac{x^2 - 2x - 1}{(x^2 + 1)(x - 1)}$$

3. $h(x) = x \ln x - x$ (appears in entropy and information measures):

$$h'(x) = \ln x + x \cdot \frac{1}{x} - 1 = \ln x + 1 - 1 = \ln x$$

So $\frac{d}{dx}[x \ln x - x] = \ln x$. **This means** $\int \ln x \, dx = x \ln x - x + C$ — a key integral formula.

4. $k(x) = \ln |\sin x|$:

$$k'(x) = \frac{\cos x}{\sin x} = \cot x$$

5. **Log-differentiation of a product:** Find $f'(x)$ where $f(x) = \frac{x^2(x+1)^3}{(2x-1)^4}$.

Take $\ln$: $\ln f = 2 \ln x + 3 \ln(x+1) - 4 \ln(2x-1)$

$$\text{Differentiate: } \frac{f'}{f} = \frac{2}{x} + \frac{3}{x+1} - \frac{8}{2x-1}$$

$$\text{So: } f'(x) = f(x) \left[\frac{2}{x} + \frac{3}{x+1} - \frac{8}{2x-1} \right]$$

4.3 Logarithmic Differentiation

When a function involves products, quotients, or powers of many terms, **logarithmic differentiation** is often far more efficient than the product/quotient rule.

Procedure: Logarithmic Differentiation

To differentiate $y = f(x)$ when f is complicated:

1. Take the natural log of both sides: $\ln y = \ln f(x)$.
2. Simplify using the laws of logarithms.
3. Differentiate both sides with respect to x : $\frac{y'}{y} = [\ln f(x)]'$.
4. Solve for $y' = y \cdot [\ln f(x)]'$.

This technique is especially powerful for **variable exponents** of the form $y = [f(x)]^{g(x)}$, which cannot be handled by the power rule alone.

Worked Example Logarithmic Differentiation

1. $y = x^x, x > 0$. (Neither pure power rule nor pure exponential rule applies.)

$$\ln y = x \ln x$$

$$\frac{y'}{y} = \ln x + x \cdot \frac{1}{x} = \ln x + 1$$

$$y' = x^x (\ln x + 1)$$

2. $y = x^{\sin x}, x > 0$:

$$\ln y = \sin x \cdot \ln x$$

$$\frac{y'}{y} = \cos x \cdot \ln x + \sin x \cdot \frac{1}{x}$$

$$y' = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$$

3. $y = \left(\frac{(x^2 + 1)^{1/2} (3x - 2)^3}{e^{2x}} \right)$:

$$\ln y = \frac{1}{2} \ln(x^2 + 1) + 3 \ln(3x - 2) - 2x$$

$$\frac{y'}{y} = \frac{x}{x^2 + 1} + \frac{9}{3x - 2} - 2$$

$$y' = y \left(\frac{x}{x^2 + 1} + \frac{9}{3x - 2} - 2 \right)$$

5 Integration of Exponential and Logarithmic Functions

Theorem Standard Integrals

$$\int e^x dx = e^x + C \quad (9)$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0 \quad (10)$$

$$\int e^{f(x)} f'(x) dx = e^{f(x)} + C \quad (11)$$

$$\int \frac{1}{x} dx = \ln |x| + C, \quad x \neq 0 \quad (12)$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C \quad (13)$$

$$\int \ln x dx = x \ln x - x + C, \quad x > 0 \quad (14)$$

$$\int a^x dx = \frac{a^x}{\ln a} + C, \quad a > 0, a \neq 1 \quad (15)$$

Worked Example Definite and Indefinite Integrals

1. $\int_0^2 e^{3x} dx$:

$$\left. \frac{1}{3} e^{3x} \right|_0^2 = \frac{e^6}{3} - \frac{1}{3} = \frac{e^6 - 1}{3} \approx 134.3$$

2. $\int \frac{2x}{x^2 + 1} dx$:

Let $u = x^2 + 1$, $du = 2x dx$: $\int \frac{du}{u} = \ln |u| + C = \ln(x^2 + 1) + C$

3. Present value of a continuous income stream $y(t) = Y_0 e^{gt}$ over $[0, T]$ at discount rate r :

$$PV = \int_0^T Y_0 e^{gt} e^{-rt} dt = Y_0 \int_0^T e^{(g-r)t} dt = \frac{Y_0}{g-r} [e^{(g-r)T} - 1], \quad g \neq r$$

If $g = r$: $PV = Y_0 T$.

4. $\int x e^{x^2} dx$:

Let $u = x^2$, $du = 2x dx$: $\frac{1}{2} \int e^u du = \frac{1}{2} e^{x^2} + C$.

5. $\int \ln x dx = x \ln x - x + C$ (use integration by parts with $u = \ln x$, $dv = dx$):

$$\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x + C.$$

6 Economic Applications

6.1 Compound Interest and Present Value

Definition Compound Interest Formulae

An initial principal P_0 at annual interest rate r :

- **Annual compounding** (once per year): $P_t = P_0(1 + r)^t$
- **m -times per year compounding**: $P_t = P_0 \left(1 + \frac{r}{m}\right)^{mt}$
- **Continuous compounding** ($m \rightarrow \infty$): $P_t = P_0 e^{rt}$

The **effective annual rate** under continuous compounding is $e^r - 1$.

The **present value** (PV) of a future amount A received in t years, with continuous discount rate r :

$$PV = Ae^{-rt}$$

Worked Example Doubling Time and Rule of 70

How long does it take for an investment to double under continuous compounding at rate r ?

$$P_0 e^{rt} = 2P_0 \Rightarrow e^{rt} = 2 \Rightarrow rt = \ln 2 \approx 0.6931$$

$$t^* = \frac{\ln 2}{r} \approx \frac{0.6931}{r}$$

Rule of 70: Since $\ln 2 \approx 0.693 \approx 70\%$, the doubling time is approximately $70/r$ (when r is expressed as a percentage).

Growth rate r	Exact doubling time $\ln 2/r$	Rule of 70 approximation
1%	69.3 years	70 years
2%	34.7 years	35 years
3%	23.1 years	23.3 years
5%	13.9 years	14 years
7%	9.9 years	10 years
10%	6.9 years	7 years

Application: GDP per capita growing at 2% doubles every ≈ 35 years; at 7% (East Asian miracle), it doubles approximately every 10 years.

Worked Example Bond Valuation Under Continuous Discounting

A zero-coupon bond pays \$1,000 in $T = 10$ years. The continuously compounded discount rate is $r = 5\% = 0.05$.

Present value: $PV = 1000 \cdot e^{-0.05 \times 10} = 1000e^{-0.5} = 1000/\sqrt{e} \approx \606.53

Duration sensitivity:

$$\frac{dPV}{dr} = 1000 \cdot (-T)e^{-rT} = -T \cdot PV = -10 \times 606.53 = -6065.3$$

A 1 percentage point rise in r (from 5% to 6%) reduces the bond price by approximately $0.01 \times 6065 = \$60.65$. This is the **modified duration** — the percentage sensitivity of bond price to yield.

More precisely: $\frac{d \ln PV}{dr} = -T = -10$, so a 1% rise in r reduces PV by approximately 10%.

6.2 Continuous Growth Models

Definition Continuous Growth

A quantity $Y(t)$ grows at constant instantaneous rate g if it satisfies the differential equation:

$$\dot{Y}(t) = g Y(t)$$

The solution is:

$$Y(t) = Y_0 e^{gt}$$

where $Y_0 = Y(0)$ is the initial value. The **instantaneous growth rate** at time t is:

$$g = \frac{d \ln Y}{dt} = \frac{\dot{Y}}{Y}$$

This is why $d \ln Y / dt$ is the natural measure of proportional growth.

Worked Example GDP Growth and the Solow Model

Suppose GDP grows at constant rate $g = 2\%$ per annum: $Y(t) = Y_0 e^{0.02t}$.

In how many years does GDP double? $t^* = \ln 2 / 0.02 \approx 34.7$ years.

Log-linearisation for estimation: $\ln Y(t) = \ln Y_0 + 0.02t$

This is linear in t — regressing $\ln Y$ on t directly estimates the growth rate g . This is the standard approach in growth accounting.

Multi-factor growth: In the Solow model, $Y = AK^\alpha L^{1-\alpha}$.

Taking logs and differentiating with respect to time:

$$\frac{d \ln Y}{dt} = \frac{d \ln A}{dt} + \alpha \frac{d \ln K}{dt} + (1 - \alpha) \frac{d \ln L}{dt}$$

$$g_Y = g_A + \alpha g_K + (1 - \alpha) g_L$$

where $g_X = \dot{X}/X$ denotes the growth rate of variable X . This is the **growth accounting equation**: output growth is the sum of TFP growth plus capital-share-weighted capital growth plus labour-share-weighted labour growth.

6.3 Elasticity via Logarithms

Definition Logarithmic Derivative and Elasticity

The **point elasticity** of y with respect to x can be written as:

$$\varepsilon_{y,x} = \frac{dy}{dx} \cdot \frac{x}{y} = \frac{d \ln y}{d \ln x}$$

The last equality follows from the chain rule: $\frac{d \ln y}{d \ln x} = \frac{dy/y}{dx/x} = \frac{dy}{dx} \cdot \frac{x}{y}$.

This means: **in a log-log regression** $\ln y = a + b \ln x$, **the coefficient b is the constant elasticity of y with respect to x .**

Worked Example Constant Elasticity Demand

Power (isoelastic) demand: $Q = AP^{-\varepsilon}$, $A, \varepsilon > 0$.

Log-linearise: $\ln Q = \ln A - \varepsilon \ln P$.

Therefore:

$$\frac{d \ln Q}{d \ln P} = -\varepsilon$$

The price elasticity is exactly $-\varepsilon$ at every price and quantity — constant along the entire demand curve. This is the **isoelastic** or **constant-elasticity** demand curve, estimated as a log-log regression.

Revenue maximisation: $R = PQ = P \cdot AP^{-\varepsilon} = AP^{1-\varepsilon}$.

$\frac{dR}{dP} = A(1-\varepsilon)P^{-\varepsilon} = 0$ requires $\varepsilon = 1$ (unit elasticity). Revenue is maximised when demand is unit elastic — consistent with the standard result $MR = P(1 - 1/\varepsilon)$.

Income elasticity: If the demand function is $Q = Bp^{-\varepsilon}Y^{\eta}$, then: $\ln Q = \ln B - \varepsilon \ln P + \eta \ln Y$.

The income elasticity is $\eta = d \ln Q / d \ln Y$, read directly from the log regression coefficient on $\ln Y$.

6.4 The Rate of Return and Continuous Compounding

Worked Example Continuous vs. Discrete Compounding

Effective annual rate (EAR):

A nominal rate of $r = 12\%$ per year compounded monthly gives:

$$\text{EAR} = \left(1 + \frac{r}{12}\right)^{12} - 1 = (1.01)^{12} - 1 \approx 12.68\%$$

Under continuous compounding:

$$\text{EAR}_{\text{continuous}} = e^r - 1 = e^{0.12} - 1 \approx 12.75\%$$

Converting between compounding conventions:

To find the continuous rate r_c equivalent to a discrete rate r_d compounded m times per year:

$$e^{r_c} = \left(1 + \frac{r_d}{m}\right)^m \implies r_c = m \ln\left(1 + \frac{r_d}{m}\right)$$

As $m \rightarrow \infty$: $r_c \rightarrow r_d$ (the nominal and continuous rates converge for high-frequency compounding).

Net Present Value under continuous discounting:

A project generates cash flows C_t at times $t_1, t_2, \dots, t_n$:

$$NPV = \sum_{i=1}^n C_{t_i} e^{-rt_i} - C_0$$

Under continuous discounting, the IRR is the r^* satisfying $NPV(r^*) = 0$, solved numerically.

6.5 Log-Linear Demand and Supply Estimation

Worked Example Structural Estimation of Demand

A researcher estimates the log-linear demand model:

$$\ln Q_d = \beta_0 + \beta_1 \ln P + \beta_2 \ln Y + \beta_3 \ln P_s + u$$

where P is own price, Y is income, P_s is substitute price, and u is an error.

Estimated results: $\hat{\beta}_0 = 4.2$, $\hat{\beta}_1 = -0.8$, $\hat{\beta}_2 = 1.2$, $\hat{\beta}_3 = 0.5$.

Interpretations:

- $\hat{\beta}_1 = -0.8$: own-price elasticity is -0.8 (inelastic demand).
- $\hat{\beta}_2 = 1.2$: income elasticity is 1.2 (luxury good, since > 1).
- $\hat{\beta}_3 = 0.5$: cross-price elasticity with the substitute is $+0.5$ (positive, confirming substitutes).

Recovering levels: From the log equation, predicted quantity at $P = 10$, $Y = 50,000$, $P_s = 8$:

$$\begin{aligned} \ln \hat{Q} &= 4.2 - 0.8 \ln 10 + 1.2 \ln 50000 + 0.5 \ln 8 \\ &= 4.2 - 0.8(2.303) + 1.2(10.820) + 0.5(2.079) = 4.2 - 1.842 + 12.984 + 1.040 = 16.382 \\ \hat{Q} &= e^{16.382} \approx 13,000,000 \text{ units} \end{aligned}$$

Note: because $e^{\hat{u}} \neq 1$, a **smearing correction** is often applied when converting log predictions to level predictions in the presence of heteroscedastic errors.

6.6 The Number e in Utility and Production

Worked Example Constant Relative Risk Aversion (CRRA) Utility

The CRRA utility function takes the form:

$$U(c) = \begin{cases} \frac{c^{1-\sigma} - 1}{1-\sigma} & \sigma \neq 1 \\ \ln c & \sigma = 1 \end{cases}$$

where $\sigma > 0$ is the coefficient of relative risk aversion.

Limiting case: As $\sigma \rightarrow 1$, the first formula approaches $\ln c$ by L'Hôpital's Rule:

$$\lim_{\sigma \rightarrow 1} \frac{c^{1-\sigma} - 1}{1-\sigma} = \lim_{\sigma \rightarrow 1} \frac{-c^{1-\sigma} \ln c}{-1} = c^0 \ln c = \ln c$$

This shows logarithmic utility is the limiting case of CRRA as $\sigma \rightarrow 1$.

Properties of $\ln c$: $U'(c) = 1/c > 0$ (positive marginal utility), $U''(c) = -1/c^2 < 0$ (diminishing MU), and the coefficient of relative risk aversion $= -cU''(c)/U'(c) = 1$ (unit RRA).

Intertemporal consumption: The Euler equation under log utility and discount factor β : $\dot{c}/c = r - \rho$ (where $\rho = -\ln \beta$ is the time preference rate). This is the **Keynes-Ramsey rule**, which is the cornerstone of intertemporal optimisation.

6.7 Depreciation and the Exponential Decay Model

Worked Example Capital Depreciation

Physical capital depreciates at constant rate $\delta > 0$. If no investment occurs, the capital stock evolves as:

$$\dot{K}(t) = -\delta K(t) \implies K(t) = K_0 e^{-\delta t}$$

Half-life of capital: $K_0 e^{-\delta t^*} = K_0/2 \implies t^* = \ln 2/\delta$.

For $\delta = 0.1$ (10% annual depreciation): $t^* = \ln 2/0.1 \approx 6.93$ years.

Value of capital stock: If equipment initially valued at \$100,000 depreciates at 15% annually: $K(t) = 100,000 \cdot e^{-0.15t}$

After 5 years: $K(5) = 100,000 \cdot e^{-0.75} \approx \$47,237$.

Straight-line vs. continuous depreciation: Under straight-line depreciation: $K(t) = K_0(1 - \delta t)$, which reaches zero at $t = 1/\delta$. Under exponential decay: $K(t) = K_0 e^{-\delta t}$, which asymptotically approaches but never reaches zero.

7 Second Derivatives and Curvature

Worked Example Concavity of $\ln x$ and Convexity of e^x

Natural logarithm:

$$(\ln x)' = \frac{1}{x} > 0, \quad (\ln x)'' = -\frac{1}{x^2} < 0 \quad \forall x > 0$$

$\ln x$ is **strictly concave** on $(0, \infty)$. This concavity underlies the **Jensen's Inequality** result in expected utility theory: for any non-degenerate random variable X ,

$$E[\ln X] < \ln E[X]$$

A risk-averse agent with log utility prefers the expected value with certainty to any lottery with the same mean — this is the mathematical foundation of risk aversion.

Natural exponential:

$$(e^x)' = e^x > 0, \quad (e^x)'' = e^x > 0 \quad \forall x \in \mathbb{R}$$

e^x is **strictly convex** and strictly increasing everywhere.

Significance for optimisation:

- The concavity of $\ln$ means that maximising $\ln f(\mathbf{x})$ is equivalent to maximising $f(\mathbf{x})$ (since $\ln$ is strictly increasing) and the $\ln$ -transformed problem has a globally concave objective whenever f is log-concave.
- Many utility maximisation problems are solved by maximising the log utility directly (Cobb-Douglas, CES), exploiting this property.

8 Practice Questions

8.1 Section A: Algebra and Laws

A1. (Laws of Exponents) Simplify each expression, writing the answer without negative or fractional exponents where possible:

(a) $\frac{x^3 \cdot x^{-2} \cdot x^{1/2}}{x^{-1}}$

(b) $(8a^6b^3)^{2/3}$

(c) $\left(\frac{4x^4y^{-2}}{9x^{-2}y^4}\right)^{1/2}$

(d) $\frac{(e^{2x})^3 \cdot e^{-4x}}{e^{x-1}}$

A2. (Laws of Logarithms) Expand or simplify each expression:

- (a) $\ln\left(\frac{x^3(x+1)^2}{\sqrt{x-1}}\right), x > 1$
- (b) $3 \ln x - \frac{1}{2} \ln(x^2 + 1) + \ln 5$
- (c) $\log_4 128$
- (d) $\ln(e^{3x} + 1) - \ln(e^{3x})$ (simplify as much as possible)

A3. (Solving Equations) Solve for x :

- (a) $e^{2x-3} = 7$
- (b) $\ln(x+2) + \ln(x-1) = \ln 4$
- (c) $5^x = 3^{x+2}$
- (d) $2 \ln x - \ln(x+1) = 0$

A4. (Compound Interest) An investor deposits \$5,000 at an annual rate of 6%.

- (a) Find the value after 10 years under annual compounding.
- (b) Find the value after 10 years under continuous compounding.
- (c) Under continuous compounding, when does the investment first exceed \$20,000?
- (d) What continuously compounded rate is equivalent to 6% compounded monthly?

8.2 Section B: Differentiation

B1. (Derivatives of Exponential Functions) Differentiate with respect to x :

- (a) $f(x) = e^{5x^2-3x+1}$
- (b) $g(x) = (3x^2 - 1)e^{-2x}$
- (c) $h(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ (hyperbolic tangent)
- (d) $k(x) = e^{e^x}$

B2. (Derivatives of Logarithmic Functions) Differentiate with respect to x :

- (a) $f(x) = \ln(x^2 + 3x + 2), x > -1$
- (b) $g(x) = x^3 \ln(2x), x > 0$
- (c) $h(x) = \ln\left(\frac{e^x + 1}{e^x - 1}\right), x > 0$
- (d) $k(x) = \ln(\ln x), x > 1$

B3. (Logarithmic Differentiation) Use logarithmic differentiation to find dy/dx :

- (a) $y = x^{\ln x}, \quad x > 0$
- (b) $y = \frac{(x^2 + 1)^3(2x - 1)^2}{(x + 3)^4}$
- (c) $y = (1 + x)^{1/x}, \quad x > 0, x \neq 0$
- (d) $y = (\sin x)^x, \quad 0 < x < \pi$

B4. (Critical Points and Optimisation) Find and classify all critical points:

- (a) $f(x) = xe^{-x^2}$
- (b) $g(x) = e^x - 3x$, find the minimum
- (c) $h(x) = x^2 \ln x - \frac{x^2}{2}, \quad x > 0$
- (d) $k(x) = \ln x - x, \quad x > 0$, find the maximum

8.3 Section C: Integration

C1. (Standard Integrals) Evaluate each integral:

- (a) $\int e^{4x-1} dx$
- (b) $\int \frac{3x^2}{x^3 + 5} dx$
- (c) $\int_1^e \frac{\ln x}{x} dx$
- (d) $\int_0^\infty e^{-rt} dt, \quad r > 0$

C2. (Present Value) A perpetuity pays C dollars per year continuously. Compute the present value at continuous discount rate $r > 0$, i.e. evaluate $PV = \int_0^\infty Ce^{-rt} dt$.

8.4 Section D: Economic Applications

D1. (Growth Accounting) GDP of country A satisfies $Y(t) = 200e^{0.04t}$ (billions of dollars, t in years from 2000).

- (a) What is GDP in 2000 and 2025?
- (b) What is the annual growth rate?
- (c) In what year does GDP first exceed \$500 billion?
- (d) What is $d(\ln Y)/dt$ and what does it measure?
- (e) Write the log-linear equation relating $\ln Y$ to t .

D2. (Elasticity) A firm faces isoelastic demand $Q = 1000P^{-1.5}Y^{0.8}$.

- (a) What is the price elasticity of demand?
- (b) What is the income elasticity?
- (c) At $P = 10$ and $Y = 1000$, what is equilibrium quantity?
- (d) If price rises by 2%, by approximately what percentage does quantity fall?
- (e) Does the firm operate on the elastic or inelastic portion of the demand curve?

D3. (Log-Linear Production) A Cobb-Douglas production function is estimated as:

$$\ln Q = 1.5 + 0.35 \ln K + 0.70 \ln L$$

- (a) What is the output elasticity of capital? Of labour?
- (b) What are the returns to scale? Justify.
- (c) If K increases by 10% and L increases by 5%, by what percentage does Q increase?
- (d) At $K = 100$ and $L = 400$, find the approximate level of Q .
- (e) Is this production function consistent with perfect competition equilibrium in the long run? Why or why not?

D4. (Optimal Advertising) A firm's profit is $\pi(A) = R_0 e^{\alpha \sqrt{A}} - A - F$, where $A > 0$ is advertising expenditure, $R_0 > 0$ is base revenue, $\alpha > 0$ is the advertising effectiveness parameter, and $F > 0$ is fixed cost.

- (a) Find $\pi'(A)$ and set up the FOC for optimal advertising A^* .
- (b) Show that the SOC $\pi''(A^*) < 0$ is satisfied.
- (c) For $R_0 = 1000$, $\alpha = 0.1$, find A^* explicitly.
- (d) Compute $d\pi^*/d\alpha$ using the Envelope Theorem and interpret.

9 Full Solutions

9.1 Solutions: Section A

Solution

A1. Laws of Exponents

$$\begin{aligned}
 \text{(a)} \quad & \frac{x^3 \cdot x^{-2} \cdot x^{1/2}}{x^{-1}} = x^{3-2+1/2+1} = x^{5/2} \\
 \text{(b)} \quad & (8a^6b^3)^{2/3} = 8^{2/3}(a^6)^{2/3}(b^3)^{2/3} = 4a^4b^2 \\
 \text{(c)} \quad & \left(\frac{4x^4y^{-2}}{9x^{-2}y^4}\right)^{1/2} = \left(\frac{4x^6}{9y^6}\right)^{1/2} = \frac{2x^3}{3y^3} = \frac{2}{3} \left(\frac{x}{y}\right)^3 \\
 \text{(d)} \quad & \frac{(e^{2x})^3 \cdot e^{-4x}}{e^{x-1}} = \frac{e^{6x} \cdot e^{-4x}}{e^x \cdot e^{-1}} = \frac{e^{2x}}{e^x \cdot e^{-1}} = e^{2x-x+1} = e^{x+1}
 \end{aligned}$$

Solution

A2. Laws of Logarithms

$$\begin{aligned}
 \text{(a)} \quad & \ln\left(\frac{x^3(x+1)^2}{\sqrt{x-1}}\right) = 3\ln x + 2\ln(x+1) - \frac{1}{2}\ln(x-1) \\
 \text{(b)} \quad & 3\ln x - \frac{1}{2}\ln(x^2+1) + \ln 5 = \ln(x^3) - \ln\sqrt{x^2+1} + \ln 5 = \ln\left(\frac{5x^3}{\sqrt{x^2+1}}\right) \\
 \text{(c)} \quad & \log_4 128 = \frac{\ln 128}{\ln 4} = \frac{7\ln 2}{2\ln 2} = \frac{7}{2} \\
 \text{Check:} \quad & 4^{7/2} = (2^2)^{7/2} = 2^7 = 128. \checkmark \\
 \text{(d)} \quad & \ln(e^{3x} + 1) - \ln(e^{3x}) = \ln\left(\frac{e^{3x} + 1}{e^{3x}}\right) = \ln(1 + e^{-3x})
 \end{aligned}$$

Solution

A3. Solving Equations

$$\begin{aligned}
 \text{(a)} \quad & e^{2x-3} = 7 \Rightarrow 2x - 3 = \ln 7 \Rightarrow x = \frac{3 + \ln 7}{2} \approx \frac{3 + 1.946}{2} \approx 2.473 \\
 \text{(b)} \quad & \ln(x+2) + \ln(x-1) = \ln 4 \Rightarrow \ln[(x+2)(x-1)] = \ln 4 \\
 & (x+2)(x-1) = 4 \Rightarrow x^2 + x - 2 = 4 \Rightarrow x^2 + x - 6 = 0 \Rightarrow (x+3)(x-2) = 0 \\
 & x = 2 \text{ (valid: } x+2 > 0 \text{ and } x-1 > 0 \text{ requires } x > 1) \text{ or } x = -3 \text{ (rejected: } \\
 & x-1 = -4 < 0). \\
 & \text{Answer: } x = 2. \\
 \text{(c)} \quad & 5^x = 3^{x+2}. \text{ Take natural logs: } x\ln 5 = (x+2)\ln 3 \Rightarrow x(\ln 5 - \ln 3) = 2\ln 3 \\
 & x = \frac{2\ln 3}{\ln 5 - \ln 3} = \frac{2\ln 3}{\ln(5/3)} \approx \frac{2(1.099)}{0.511} \approx 4.30 \\
 \text{(d)} \quad & 2\ln x - \ln(x+1) = 0 \Rightarrow \ln(x^2) = \ln(x+1) \Rightarrow x^2 = x+1 \\
 & x^2 - x - 1 = 0 \Rightarrow x = \frac{1 + \sqrt{5}}{2} \approx 1.618 \text{ (taking positive root; } x \text{ must be } > 0). \\
 & \text{This is the golden ratio } \phi!
 \end{aligned}$$

Solution

A4. Compound Interest

$$\text{(a) Annual compounding: } A = 5000(1.06)^{10} \approx 5000(1.7908) \approx \$8,954$$

(b) Continuous compounding: $A = 5000e^{0.06 \times 10} = 5000e^{0.6} \approx 5000(1.8221) \approx \$9,111$

Continuous compounding yields $\approx \$157$ more than annual compounding.

(c) $5000e^{0.06t} > 20000 \Rightarrow e^{0.06t} > 4 \Rightarrow 0.06t > \ln 4 \Rightarrow t > \frac{\ln 4}{0.06} = \frac{1.386}{0.06} \approx 23.1$ years.

(d) Monthly compounding at $r = 6\%$: $(1 + 0.06/12)^{12} = (1.005)^{12} \approx e^{r_c}$
 $r_c = 12 \ln(1.005) \approx 12(0.004988) \approx 5.986\%$

So the equivalent continuous rate is $\approx 5.99\%$, slightly less than the nominal 6%.

9.2 Solutions: Section B

Solution

B1. Derivatives of Exponential Functions

(a) $f'(x) = e^{5x^2-3x+1}(10x-3)$

(b) Product rule, $u = 3x^2 - 1, v = e^{-2x}$:

$$g'(x) = 6x e^{-2x} + (3x^2 - 1)(-2)e^{-2x} = e^{-2x}(6x - 6x^2 + 2) = 2e^{-2x}(3x - 3x^2 + 1)$$

(c) Quotient rule, $u = e^x - e^{-x}, v = e^x + e^{-x}$:

$$h'(x) = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

Numerator: $(e^x + e^{-x})^2 - (e^x - e^{-x})^2 = 4e^x e^{-x} = 4$ (difference of squares).

So $h'(x) = \frac{4}{(e^x + e^{-x})^2} = \text{sech}^2(x)$. This is the derivative of $\tanh(x)$.

(d) Chain rule twice: $k'(x) = e^{e^x} \cdot e^x = e^x e^{e^x}$

Solution

B2. Derivatives of Logarithmic Functions

(a) $f'(x) = \frac{2x+3}{x^2+3x+2}$

(b) Product rule, $u = x^3, v = \ln(2x)$:

$$g'(x) = 3x^2 \ln(2x) + x^3 \cdot \frac{1}{x} = 3x^2 \ln(2x) + x^2 = x^2(3 \ln(2x) + 1)$$

(c) $h(x) = \ln(e^x + 1) - \ln(e^x - 1), x > 0$:

$$h'(x) = \frac{e^x}{e^x + 1} - \frac{e^x}{e^x - 1} = e^x \left(\frac{1}{e^x + 1} - \frac{1}{e^x - 1} \right) = e^x \cdot \frac{(e^x - 1) - (e^x + 1)}{(e^x + 1)(e^x - 1)} = \frac{-2e^x}{e^{2x} - 1}$$

(d) Let $u = \ln x, y = \ln u$:

$$k'(x) = \frac{1}{\ln x} \cdot \frac{1}{x} = \frac{1}{x \ln x}$$

Solution

B3. Logarithmic Differentiation

(a) $y = x^{\ln x}$, $x > 0$. $\ln y = (\ln x)^2$.

$$\frac{y'}{y} = 2 \ln x \cdot \frac{1}{x} \quad y' = x^{\ln x} \cdot \frac{2 \ln x}{x}$$

(b) $\ln y = 3 \ln(x^2 + 1) + 2 \ln(2x - 1) - 4 \ln(x + 3)$

$$\frac{y'}{y} = \frac{6x}{x^2 + 1} + \frac{4}{2x - 1} - \frac{4}{x + 3}$$

$$y' = y \left(\frac{6x}{x^2 + 1} + \frac{4}{2x - 1} - \frac{4}{x + 3} \right)$$

(c) $y = (1 + x)^{1/x}$. $\ln y = \frac{1}{x} \ln(1 + x)$.

$$\frac{y'}{y} = -\frac{1}{x^2} \ln(1 + x) + \frac{1}{x} \cdot \frac{1}{1 + x} = \frac{1}{x} \left[\frac{1}{1 + x} - \frac{\ln(1 + x)}{x} \right]$$

$$y' = (1 + x)^{1/x} \cdot \frac{1}{x} \left[\frac{1}{1 + x} - \frac{\ln(1 + x)}{x} \right]$$

Note: as $x \rightarrow 0^+$, $y \rightarrow e$ (this is the definition-of- e limit in disguise).

(d) $y = (\sin x)^x$. $\ln y = x \ln \sin x$.

$$\frac{y'}{y} = \ln \sin x + x \cdot \frac{\cos x}{\sin x} = \ln \sin x + x \cot x$$

$$y' = (\sin x)^x (\ln \sin x + x \cot x)$$

Solution

B4. Critical Points

(a) $f(x) = xe^{-x^2}$. $f'(x) = e^{-x^2} + x(-2x)e^{-x^2} = e^{-x^2}(1 - 2x^2) = 0$.

Since $e^{-x^2} > 0$: $1 - 2x^2 = 0 \Rightarrow x = \pm 1/\sqrt{2}$.

$$f''(x) = -2xe^{-x^2}(1 - 2x^2) + e^{-x^2}(-4x) = e^{-x^2}(-2x + 4x^3 - 4x) = e^{-x^2}(4x^3 - 6x)$$

$$f''(1/\sqrt{2}) = e^{-1/2}(4/(2\sqrt{2}) - 6/\sqrt{2}) = e^{-1/2}(2/\sqrt{2} - 6/\sqrt{2}) = -4/(\sqrt{2}e^{1/2}) < 0: \text{local max.}$$

$$f''(-1/\sqrt{2}) > 0: \text{local min.}$$

$$f(1/\sqrt{2}) = (1/\sqrt{2})e^{-1/2} = 1/(\sqrt{2}\sqrt{e}) \approx 0.429 \text{ (local max).}$$

$$f(-1/\sqrt{2}) = -1/(\sqrt{2}\sqrt{e}) \approx -0.429 \text{ (local min).}$$

(b) $g(x) = e^x - 3x$. $g'(x) = e^x - 3 = 0 \Rightarrow e^x = 3 \Rightarrow x^* = \ln 3$.

$g''(x) = e^x > 0$ everywhere, so g is strictly convex: $x^* = \ln 3$ is the **global minimum**.

$$g(\ln 3) = e^{\ln 3} - 3 \ln 3 = 3 - 3 \ln 3 = 3(1 - \ln 3) \approx 3(1 - 1.099) \approx -0.296$$

(c) $h(x) = x^2 \ln x - x^2/2$. $h'(x) = 2x \ln x + x - x = 2x \ln x = 0$.

$2x \ln x = 0 \Rightarrow x = 0$ (not in domain $x > 0$) or $\ln x = 0 \Rightarrow x = 1$.

$h''(x) = 2 \ln x + 2$. $h''(1) = 0 + 2 = 2 > 0$: local (and global) **minimum**.

$$h(1) = 0 - 1/2 = -1/2.$$

(d) $k(x) = \ln x - x$. $k'(x) = 1/x - 1 = 0 \Rightarrow x = 1$.

$k''(x) = -1/x^2$. $k''(1) = -1 < 0$: **global maximum** (since k is strictly concave).

$$k(1) = \ln 1 - 1 = 0 - 1 = -1.$$

Note: $\ln x < x$ for all $x > 0$, i.e. $k(x) = \ln x - x < -1 + (-1) = -1$... actually $k(1) = \ln 1 - 1 = -1$ and $k(x) < k(1) = -1$ for all $x \neq 1$ confirms this is a

maximum with maximum value -1 .
Equivalently this proves $\ln x \leq x - 1$ for all $x > 0$ (with equality at $x = 1$) — a fundamental inequality.

9.3 Solutions: Section C

Solution

C1. Standard Integrals

(a) $\int e^{4x-1} dx = \frac{1}{4}e^{4x-1} + C$

(b) $\int \frac{3x^2}{x^3+5} dx$: let $u = x^3 + 5$, $du = 3x^2 dx$: $\int \frac{du}{u} = \ln|u| + C = \ln(x^3 + 5) + C$
(since $x^3 + 5 > 0$)

(c) $\int_1^e \frac{\ln x}{x} dx$: let $u = \ln x$, $du = dx/x$. Limits: $x = 1 \Rightarrow u = 0$; $x = e \Rightarrow u = 1$.

$$\int_0^1 u du = \frac{u^2}{2} \Big|_0^1 = \frac{1}{2}$$

(d) $\int_0^\infty e^{-rt} dt = -\frac{1}{r}e^{-rt} \Big|_0^\infty = 0 - \left(-\frac{1}{r}\right) = \frac{1}{r}$

(Valid since $e^{-rt} \rightarrow 0$ as $t \rightarrow \infty$ for $r > 0$.)

Solution

C2. Present Value of a Perpetuity

$$PV = \int_0^\infty Ce^{-rt} dt = C \int_0^\infty e^{-rt} dt = C \cdot \frac{1}{r} = \frac{C}{r}$$

This is the **Gordon perpetuity formula**: the present value of a perpetual cash flow C at discount rate r is simply C/r .

Example: A bond paying \$100 per year in perpetuity at $r = 5\%$ is worth $100/0.05 = \$2,000$.

Growing perpetuity: If the cash flow grows at rate $g < r$:

$$PV = \int_0^\infty Ce^{gt}e^{-rt} dt = C \int_0^\infty e^{-(r-g)t} dt = \frac{C}{r-g}$$

This is the **Gordon Growth Model** in continuous time, fundamental to equity valuation.

9.4 Solutions: Section D

Solution

D1. Growth Accounting

$Y(t) = 200e^{0.04t}$, $t = 0$ corresponds to year 2000.

(a) $Y(0) = 200$ billion (year 2000); $Y(25) = 200e^1 = 200e \approx \543.7 billion (year 2025).

- (b) Annual growth rate = 4% (the coefficient on t in the exponent).
- (c) $200e^{0.04t} > 500 \Rightarrow e^{0.04t} > 2.5 \Rightarrow 0.04t > \ln 2.5 \Rightarrow t > \frac{\ln 2.5}{0.04} \approx \frac{0.916}{0.04} \approx 22.9$ years.
Year: $2000 + 22.9 \approx 2023$.
- (d) $\frac{d \ln Y}{dt} = 0.04$. This is the **instantaneous proportional growth rate** of GDP — the continuous-time analogue of the percentage growth rate. It equals the log-derivative of Y with respect to time.
- (e) $\ln Y = \ln 200 + 0.04t$, a linear function of time t . The slope 0.04 is the continuously compounded growth rate of GDP.

Solution

D2. Elasticity

$$Q = 1000P^{-1.5}Y^{0.8}.$$

- (a) Price elasticity = -1.5 (constant, read directly from the exponent on P).
- (b) Income elasticity = 0.8 (constant, read from the exponent on Y).
- (c) $Q = 1000(10)^{-1.5}(1000)^{0.8} = 1000 \cdot 10^{-3/2} \cdot 10^{2.4}$
 $= 1000 \cdot 10^{-1.5+2.4} = 1000 \cdot 10^{0.9} \approx 1000 \times 7.943 \approx 7,943$ units.
- (d) Price elasticity = -1.5 . A 2% rise in P reduces Q by approximately $1.5 \times 2\% = 3\%$.
- (e) $|\varepsilon_{Q,P}| = 1.5 > 1$: the firm operates on the **elastic** portion of the demand curve. A price increase reduces total revenue (since revenue falls when demand is elastic). To maximise revenue, the firm should reduce price until $|\varepsilon| = 1$.

Solution

D3. Log-Linear Production

$$\ln Q = 1.5 + 0.35 \ln K + 0.70 \ln L$$

- (a) Output elasticity of capital = 0.35; of labour = 0.70.
- (b) Returns to scale = $0.35 + 0.70 = 1.05 > 1$: **increasing returns to scale**. A 1% increase in all inputs raises output by 1.05%.
- (c) With $\Delta \ln K \approx \Delta K/K = 10\% = 0.10$ and $\Delta \ln L \approx 0.05$:

$$\Delta \ln Q \approx 0.35(0.10) + 0.70(0.05) = 0.035 + 0.035 = 0.07$$

Output increases by approximately 7%.

$$\begin{aligned} \text{(d)} \ln Q &= 1.5 + 0.35 \ln 100 + 0.70 \ln 400 \\ &= 1.5 + 0.35(4.605) + 0.70(5.991) \\ &= 1.5 + 1.612 + 4.194 = 7.306 \\ Q &= e^{7.306} \approx 1,494 \text{ units.} \end{aligned}$$

- (e) Under IRS ($\alpha + \beta = 1.05 > 1$), if factors are paid their marginal products, total payments = $(\alpha + \beta)Q = 1.05Q > Q$. Total factor payments exceed output, implying negative profits. This is inconsistent with long-run competitive equilibrium where zero profit requires $\alpha + \beta = 1$ (CRS). The firm would either exhibit natural monopoly tendencies or the IRS estimate is due to measurement error/omitted variables.

Solution

D4. Optimal Advertising

$$\pi(A) = R_0 e^{\alpha\sqrt{A}} - A - F.$$

$$(a) \pi'(A) = R_0 e^{\alpha\sqrt{A}} \cdot \frac{\alpha}{2\sqrt{A}} - 1 = 0$$

$$\text{FOC: } \frac{\alpha R_0 e^{\alpha\sqrt{A}}}{2\sqrt{A}} = 1$$

$$(b) \pi''(A) = R_0 e^{\alpha\sqrt{A}} \left[\left(\frac{\alpha}{2\sqrt{A}} \right)^2 - \frac{\alpha}{4A^{3/2}} \right]$$

$$= R_0 e^{\alpha\sqrt{A}} \left[\frac{\alpha^2}{4A} - \frac{\alpha}{4A^{3/2}} \right]$$

$$= \frac{R_0 e^{\alpha\sqrt{A}}}{4A} \left[\alpha^2 - \frac{\alpha}{\sqrt{A}} \right]$$

$$= \frac{R_0 \alpha e^{\alpha\sqrt{A}}}{4A} \left[\alpha - \frac{1}{\sqrt{A}} \right]$$

$$\text{At the FOC, } \frac{\alpha R_0 e^{\alpha\sqrt{A}}}{2\sqrt{A}} = 1, \text{ so } R_0 e^{\alpha\sqrt{A}} = \frac{2\sqrt{A}}{\alpha}.$$

$$\text{Substituting: } \pi''(A^*) = \frac{2\sqrt{A^*}}{4A^* \alpha^{-1} \cdot \alpha} \left(\alpha - \frac{1}{\sqrt{A^*}} \right) = \frac{1}{2\sqrt{A^*}} \left(\alpha - \frac{1}{\sqrt{A^*}} \right)$$

For the SOC, we need $\pi''(A^*) < 0$, which requires $\alpha < 1/\sqrt{A^*}$, i.e. $\sqrt{A^*} < 1/\alpha$, i.e. $A^* < 1/\alpha^2$.

From the FOC: $\frac{\alpha R_0 e^{\alpha\sqrt{A^*}}}{2\sqrt{A^*}} = 1$. For small α , $e^{\alpha\sqrt{A^*}} \approx 1 + \alpha\sqrt{A^*}$, giving $\sqrt{A^*} \approx \frac{\alpha R_0}{2}$ for small values. Since $R_0 > 0$ and $\alpha > 0$, for sufficiently large R_0 the interior maximum exists and is confirmed by $\pi'' < 0$.

$$(c) \text{ With } R_0 = 1000, \alpha = 0.1. \text{ FOC: } \frac{0.1 \times 1000 \times e^{0.1\sqrt{A}}}{2\sqrt{A}} = 1$$

$$50 \frac{e^{0.1\sqrt{A}}}{\sqrt{A}} = 1$$

Let $u = \sqrt{A}$, so $50 \frac{e^{0.1u}}{u} = 1$, i.e. $50e^{0.1u} = u$.

This is a transcendental equation. We solve numerically.

At $u = 50$: $50e^5 \approx 50 \times 148.4 = 7420 \neq 50$.

At $u = 200$: $50e^{20}$ is enormous. The equation $50e^{0.1u} = u$ has the exponential growing faster, so we need u large. Actually: rearranging, $e^{0.1u}/u = 1/50$. For large u , $e^{0.1u}$ dominates u , so the product $e^{0.1u}/u \rightarrow \infty$. The equation must be solved numerically; for $R_0 = 1000, \alpha = 0.1$, we find $A^* \approx (41.3)^2 \approx 1706$ by iterating.

$$(d) \text{ Envelope Theorem: } \pi^*(A; \alpha) = R_0 e^{\alpha\sqrt{A^*}} - A^* - F.$$

$$\frac{d\pi^*}{d\alpha} = \frac{\partial \pi}{\partial \alpha} \Big|_{A=A^*} = R_0 e^{\alpha\sqrt{A^*}} \cdot \sqrt{A^*} = \sqrt{A^*} \cdot R_0 e^{\alpha\sqrt{A^*}}$$

Since $R_0 e^{\alpha\sqrt{A^*}} = 2\sqrt{A^*}/\alpha$ (from the FOC):

$$\frac{d\pi^*}{d\alpha} = \sqrt{A^*} \cdot \frac{2\sqrt{A^*}}{\alpha} = \frac{2A^*}{\alpha}$$

Interpretation: a unit increase in advertising effectiveness α raises maximum profit by $2A^*/\alpha$. A firm with highly effective advertising (large α) is more sensitive to changes in α , and profits increase more than proportionally with advertising effectiveness.

Summary: Key Formulae and Results

Result	Formula / Statement
Definition of e	$e = \lim_{n \rightarrow \infty} (1 + 1/n)^n = \sum_{n=0}^{\infty} 1/n! \approx 2.71828$
Continuous compounding	$P_t = P_0 e^{rt}; PV = Ae^{-rt}$
Doubling time	$t^* = \ln 2/r \approx 0.693/r$ (Rule of 70: $\approx 70/r\%$)
Derivative of e^x	$(e^x)' = e^x; (e^{f(x)})' = f'(x)e^{f(x)}$
Derivative of a^x	$(a^x)' = a^x \ln a; (a^{f(x)})' = f'(x) \ln a \cdot a^{f(x)}$
Derivative of $\ln x$	$(\ln x)' = 1/x; (\ln f(x))' = f'(x)/f(x)$
Log differentiation	Take $\ln$, differentiate, solve for $y' = y \cdot (\ln y)'$
Integral of e^{ax}	$\int e^{ax} dx = (1/a)e^{ax} + C$
Integral of $1/x$	$\int (1/x) dx = \ln x + C; \int (f'/f) dx = \ln f + C$
Integral of $\ln x$	$\int \ln x dx = x \ln x - x + C$
Log laws	$\ln(MN) = \ln M + \ln N; \ln(M/N) = \ln M - \ln N; \ln(M^p) = p \ln M$
Elasticity via log	$\varepsilon_{y,x} = d \ln y / d \ln x$; log-log regression gives constant elasticity
Growth rate	$g = d \ln Y / dt = \dot{Y}/Y; Y(t) = Y_0 e^{gt}$
Jensen's inequality	$E[\ln X] \leq \ln E[X]$ (concavity of $\ln$; basis of risk aversion)
Key inequality	$\ln x \leq x - 1$ for all $x > 0$ (equality at $x = 1$)
Perpetuity PV	$\int_0^{\infty} Ce^{-rt} dt = C/r$; growing: $C/(r - g)$
CRRA utility	$U(c) = \ln c$ when $\sigma = 1$; limit of $(c^{1-\sigma} - 1)/(1 - \sigma)$

References

- **Primary:** Chiang & Wainwright, *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw-Hill, 2005. Chapter 10.
- **Supplementary:** Hoy et al., *Mathematics for Economics*, 3rd ed., MIT Press, 2011. Chapter 6.
- **Supplementary:** Simon & Blume, *Mathematics for Economists*, Norton, 1994. Chapter 25.

End of Notes — ECON 320 Exponential and Logarithmic Functions — May 2026